

Computational Microelectronics

Lecture 10 (Pre-recorded)

Sung-Min Hong (smhong@gist.ac.kr)

Semiconductor Device Simulation Laboratory

Department of Electrical Engineering and Computer Science

Gwangju Institute of Science and Technology (GIST)

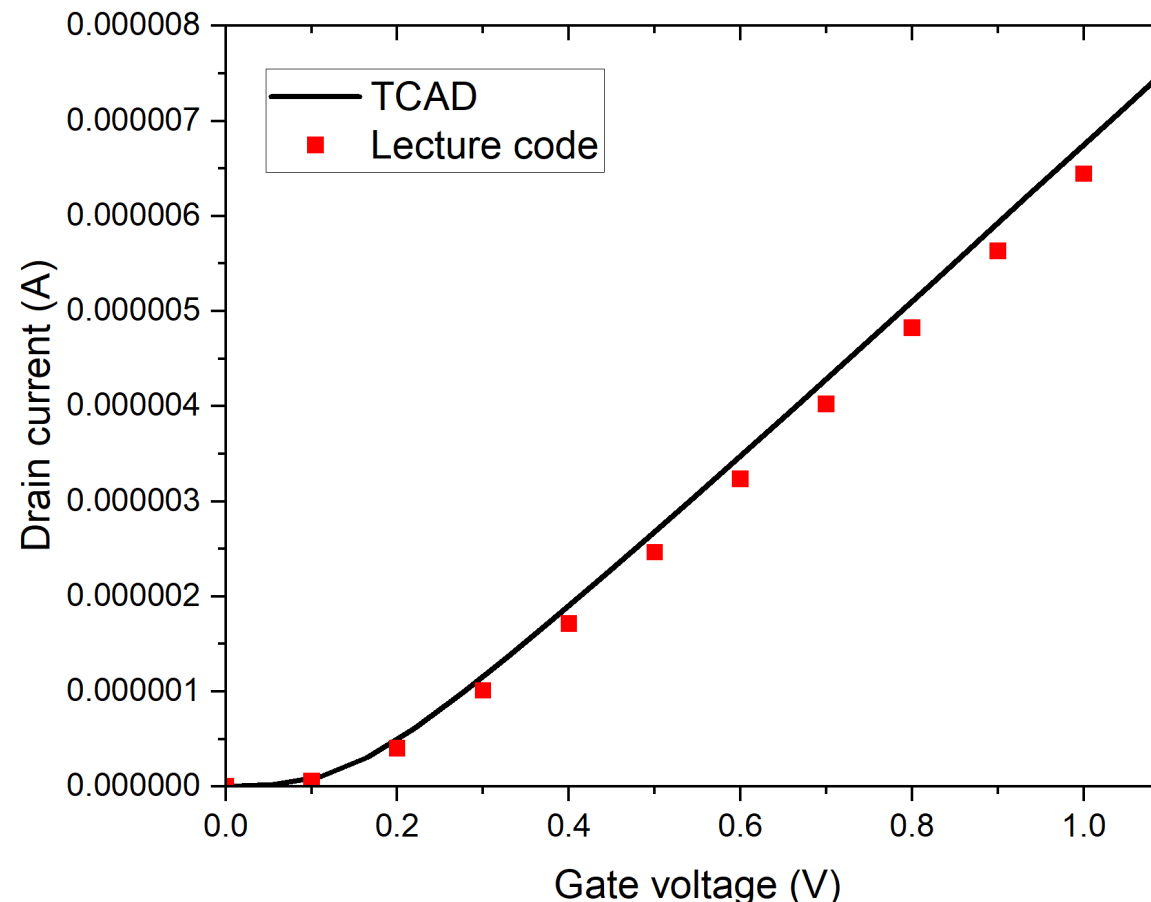
L10

Issues

- We have written a code to calculate I_D .
 - ~~1) Electron continuity equation is analytically solved.~~
 - ~~2) MOS equation and continuity equation are separately treated.~~
 - 3) Mobility is assumed to be constant.
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

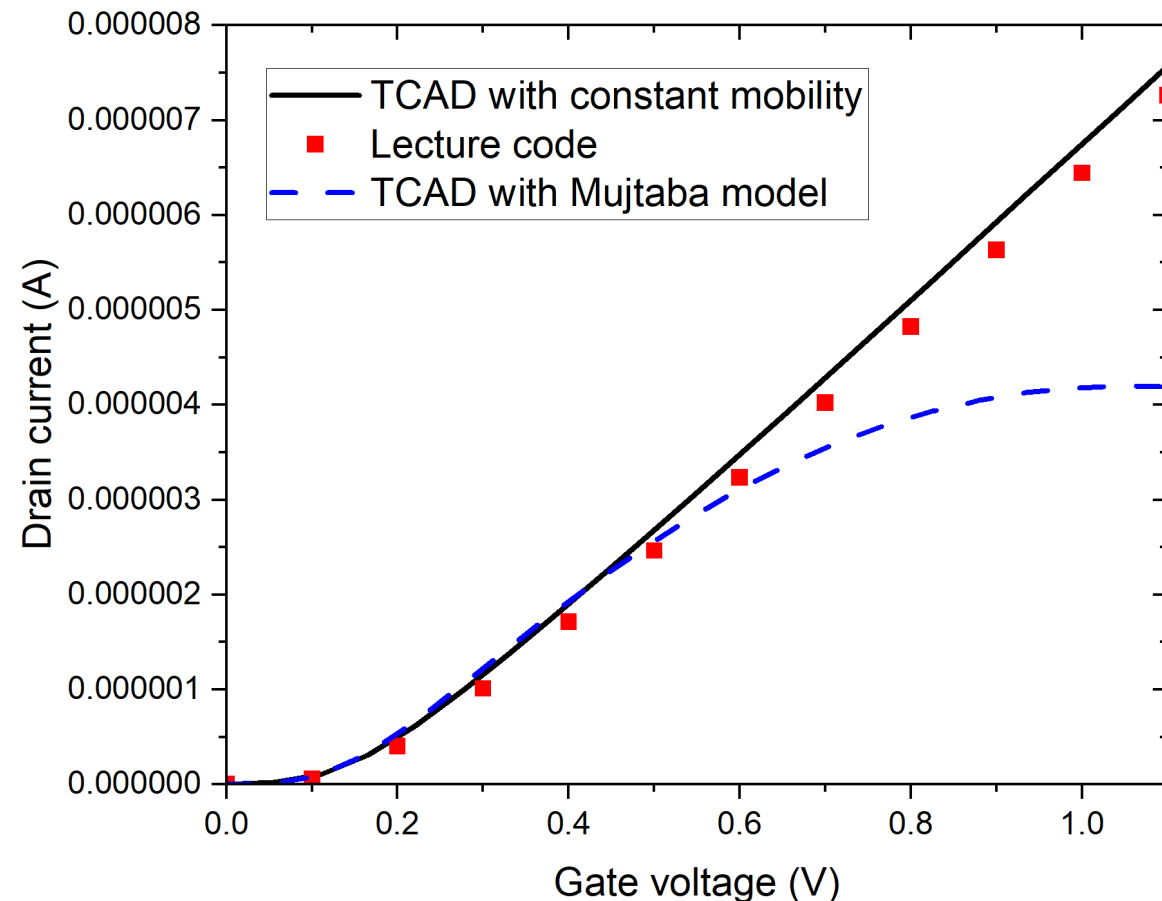
$300 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (Also, $\chi = 4.0727 \text{ eV}$)

- $I_D - V_G$ curve at a low V_D of 0.01 V
 - A constant mobility of $300 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ is used.



Mobility model is (partially) turned on.

- Mobility model by Mujtaba (for inversion layers)
 - A huge difference at high V_G values
 - The mobility varies.
 - Consider $\mu_n(x_{i+0.5})$.



Implementation

- Now, the electron continuity equation should include the mobility.

$$\begin{aligned} F_{\phi_{n,i}} &= Q_{n,i} \mu_{n,i+0.5} \phi_{n,i+1} - (Q_{n,i} \mu_{n,i+0.5} + Q_{n,i-1} \mu_{n,i-0.5}) \phi_{n,i} \\ &+ Q_{n,i-1} \mu_{n,i-0.5} \phi_{n,i-1} = 0 \end{aligned}$$

– In terms of μ_n ,

$$\begin{aligned} F_{\phi_{n,i}} &= \mu_{n,i+0.5} Q_{n,i} (\phi_{n,i+1} - \phi_{n,i}) - \mu_{n,i-0.5} Q_{n,i-1} (\phi_{n,i} - \phi_{n,i-1}) \\ &= 0 \end{aligned}$$

– (Depending on the exact form of μ_n , the Jacobian matrix may differ.)

The Mujtaba model is very complicated.

- Let us consider its surface roughness (SR) component only.

$$\frac{1}{\mu_n} = \frac{1}{\mu_n^{other\ mechanisms}} + \frac{1}{\mu_n^{SR}}$$

($\mu_n^{other\ mechanisms}$ is $600\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ in the next slides.)

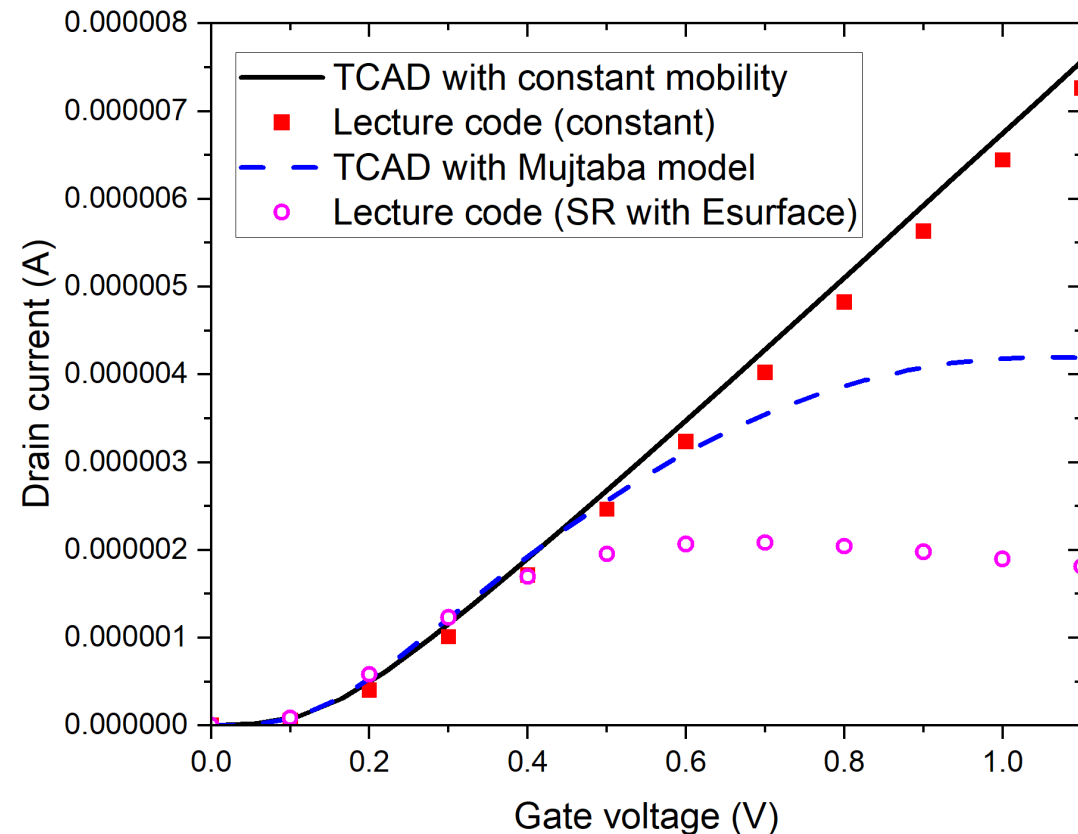
- Surface-roughness-limited mobility, μ_n^{SR}

$$\frac{1}{\mu_n^{SR}} = \frac{(E_{normal}/E_0)^2}{\text{mujtaba. delta}}$$

- Here, the default value of mujtaba. delta is $7.15 \times 10^{14}\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$.
- The normalization constant E_0 is 1 V cm^{-1} .
- Then, what is the normal electric field, E_{normal} ?

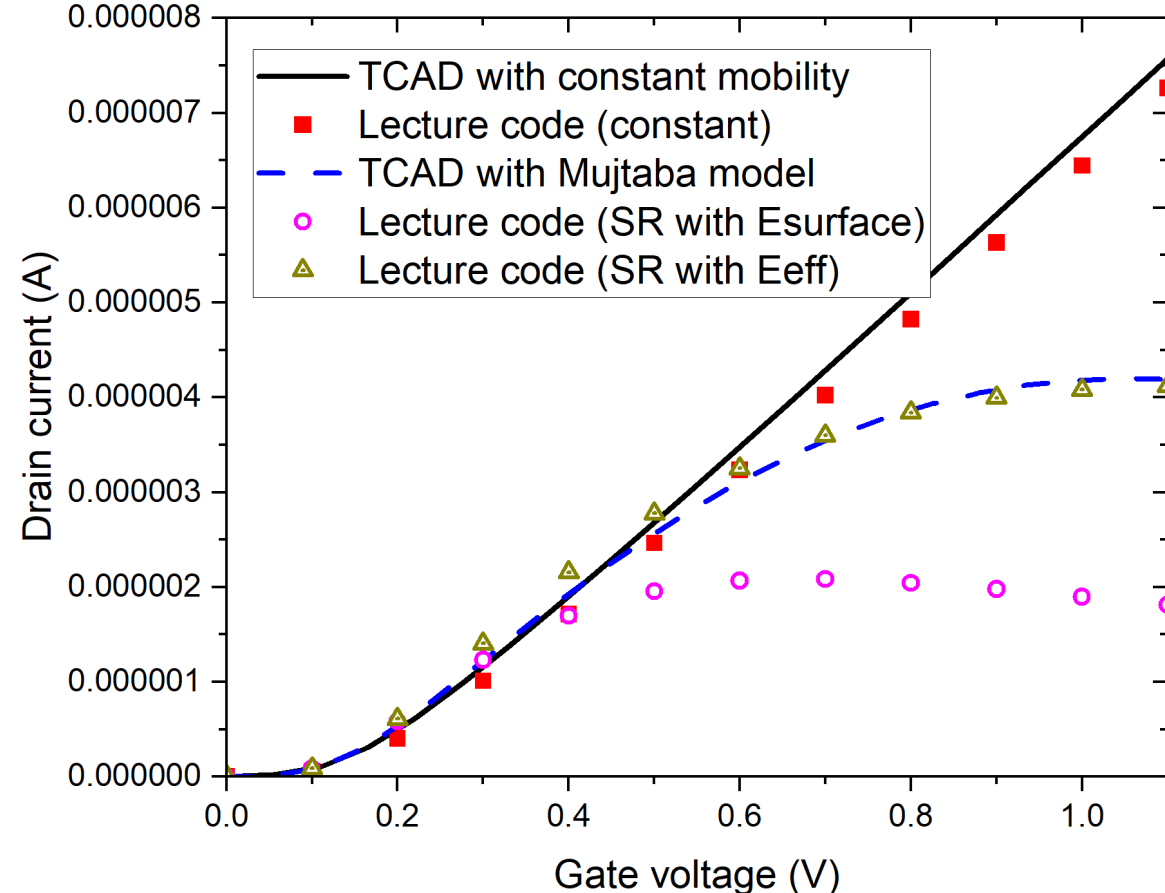
If we have the z -directional mesh,

- The normal electric field can be calculated at each node.
 - On the other hand, we do not consider the z -directional mesh.
 - Why don't we use the surface electric field, $E_z(z = 0)$?
 - The mobility reduction is overestimated.



A typical way

- The average electric field is not the surface electric field.
 - Effective electric field, $E_{eff} = \frac{1}{\epsilon_{si}} \left(|Q_d| + \frac{1}{2} |Q_n| \right)$
 - Increased mobility



$$\frac{\partial \mu_n}{\partial E_{normal}} ?$$

- From $\frac{1}{\mu_n} = \frac{1}{\mu_n^{other mechanisms}} + \frac{1}{\mu_n^{SR}}$ and $\frac{1}{\mu_n^{SR}} = \frac{(E_{normal}/E_0)^2}{\text{mujtaba.delta}}$,

– It is found that

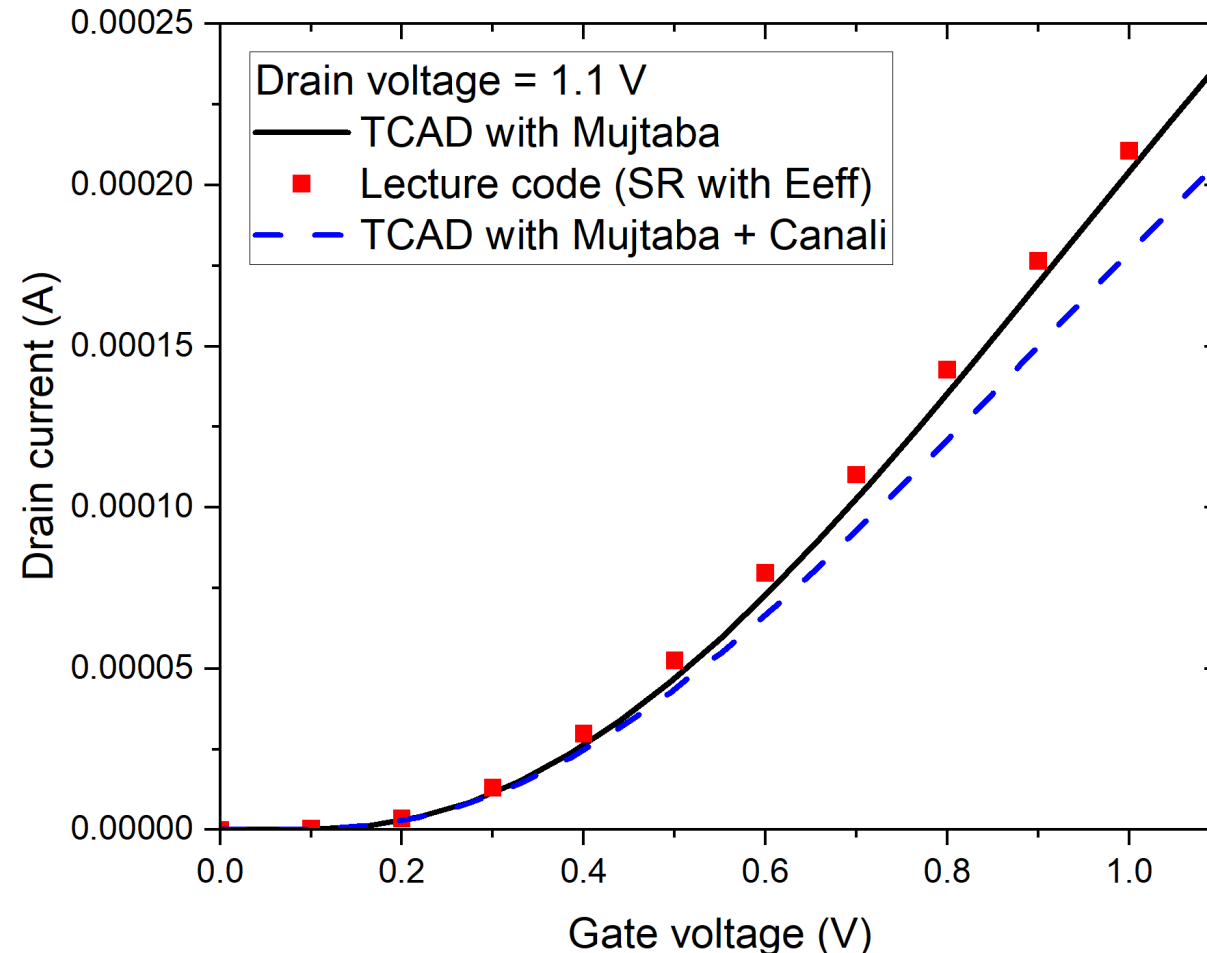
$$\frac{\partial \mu_n}{\partial E_{normal}} = -\mu_n^2 \frac{2(E_{normal}/E_0^2)}{\text{mujtaba.delta}}$$

– Furthermore, we know $\frac{\partial Q_s}{\partial \phi_s}$, $\frac{\partial Q_s}{\partial \phi_n}$, $\frac{\partial Q_n}{\partial \phi_s}$, and $\frac{\partial Q_n}{\partial \phi_n}$. When E_{normal} is written in terms of Q_s and Q_n , its dependence can be evaluated accurately.

– Quantities such as $\frac{\partial \mu_n}{\partial \phi_s}$ can be calculated.

Velocity saturation (Canali model)

- The impact of velocity saturation is marginal in a long-channel MOSFET.



Thank you!